**Group Members:**

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_____

# Innovative Product Development

**Title:** Learnova — AI-powered redesign of outdated lecture slides into illustrated, interactive teaching decks

Domain: Education Technology · Artificial Intelligence · Document Understanding

Group No. 1 · Department of Artificial Intelligence and Data Science, DJSCE · Interdisciplinary Project Review, Odd Sem 2026

# IDEA & PROPOSED SOLUTION

*From text walls to teaching decks — without losing a single point the teacher wrote*

**LEARNOVA · Group 1**
IPD Review · AI & DS, DJSCE

**In five real course decks (306 slides) we found 83 untitled slides, equations pasted as pictures, and not one diagram, quiz or interactive element**

**Proposed solution:** Faculty reuse the same slide decks for years, and students revise from them before every examination. Learnova is a web platform that takes the deck a teacher already has — a PowerPoint, a slide PDF, Word notes or a typed topic — and rebuilds it as an **editable PowerPoint, an interactive web deck and a study pack**. Every point of the original is kept; what changes is how it is shown: a visual for every idea, an illustration on every slide, and quizzes, solved problems and labs that let students work with the material instead of only reading it.

## Detailed explanation of the solution

- **Slide reader:** PDF/PPTX read page by page — titles, bullet levels, tables, figures; OCR for picture-only slides.
- **Decision engine:** 27 detectors choose process, timeline, comparison, definition, chart or cards — with a reason.
- **Assertion titles:** every title states the slide's point (AI with key, rules without).
- **Illustrations:** AI-written diagram, teacher's figure or offline scene on every slide, each with an HTML link.
- **Interaction:** checkpoint quizzes, exact step-by-step solvers, labs for NLP, ML and IPCV topics.
- **Outputs:** editable PPTX, Reveal.js web deck, flashcards, Anki, quiz bank, share link, tutor.

## How it addresses the problem

- **Text walls →** one idea per slide, drawn as a visual at a readable density.
- **Untitled slides →** 83 → 0; split parts are named after what they cover.
- **Pictures of equations →** kept as full-size figures; never replaced by OCR noise.
- **Passive reading →** quizzes, steppers and labs on the slide itself.
- **Hours of redesign →** a finished deck in about 10–60 s, fully editable.
- **Unreliable AI →** source text kept verbatim; 98–100% of facts retained.

## Innovation and uniqueness

- **Redesign, not generate:** starts from the teacher's deck; Gamma, Copilot and Canva start from a prompt.
- **Faithful by construction:** every AI output is checked against the slide; numbers are computed.
- **Explainable design:** each visual choice comes with its rule and confidence.
- **Glyph-measured layout:** no overflow and no word ever broken across lines.
- **Two fallbacks:** 5-key Gemini pool → Groq/NVIDIA → full offline logic.
- **Built-in labs:** Region Growing, Split & Merge, CYK, Viterbi, NB-WSD, HMM tagging.

**PHASE 1 · Sep–Dec 2025**

**Research & design**
Literature survey, competitor study, problem statement, architecture, UI plan

**PHASE 2 · Jan–Mar 2026**

**Prototype V1**
Parsing, AI rewriting, engagement score, quiz generation, Streamlit demo

**PHASE 3 · Apr–Jul 2026**

**Learnova V2 platform**
FastAPI + React, decision engine, PPTX & web builders, OTP accounts, library

**PHASE 4 · Aug–Oct 2026**

**Real lecture decks**
Slide reader + OCR, illustrations, explainers, labs, AI keys, papers

**LIVE CODE**
Full source, 755 tests, papers and demo decks on GitHub
**SDP42/LearnovaV2**

# LITERATURE REVIEW / EXISTING SYSTEMS

*What research and today's tools do — and the gap Learnova fills*

**LEARNOVA · Group 1**
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| # | Reference / existing system | Method & key findings | Gap / scope for improvement |
|---|---|---|---|
| 1 | Hu & Wan, "PPSGen", IEEE TKDE 27(4), 2015 | Learns sentence importance with regression; ILP selects sentences into slides for academic papers. | Papers only; summarises so content is lost; no visuals; cannot redesign existing slides. |
| 2 | Fu et al., "DOC2PPT", AAAI 36, 2022 | Hierarchical neural summariser + figure selection + layout prediction from paper–slide pairs. | Needs training pairs; abstractive text can alter facts; input must be a scientific paper. |
| 3 | Sun et al., "D2S", NAACL 2021 | Slide titles used as queries; retrieval + QA-style summarisation fills each slide. | Text only; no figures, quizzes or interaction; not for an existing lecture deck. |
| 4 | Maynez et al., "Faithfulness in abstractive summarization", ACL 2020 | Abstractive models hallucinate and omit content in a large share of outputs. | Motivates keeping source text verbatim and checking every AI output. |
| 5 | Garner & Alley, assertion–evidence, IJEE 29(6), 2013; Mayer, Multimedia Learning, 2009 | Sentence headlines + visual evidence beat bullet lists; one idea per slide, words near pictures. | Design guidance only — no tool applies it automatically to old decks. |
| 6 | Sweller & Cooper 1985; Hundhausen et al. 2002 (worked examples, algorithm visualisation) | Studying solved steps and interacting with a visual simulation improves learning. | Lecture slides show finished answers; no step-by-step or interactive layer. |
| 7 | Gamma, MS Copilot in PowerPoint, Gemini in Slides, Canva Magic Design (2026) | LLM writes an outline, then fills a template; attractive new decks from a prompt. | Rewrite/merge the teacher's points, drop figures, no labs/solvers, need internet & paid plans. |

## Gap → how Learnova closes it

| | |
|---|---|
| Generates new decks | Redesigns the existing deck |
| Paraphrases, drops items | Verbatim content + fact check |
| Loses figures/equations | Keeps every figure; OCR only when reliable |
| No visuals by meaning | 27-detector decision engine |
| No illustration | AI diagram / own figure / offline scene |
| No interaction | Quizzes, solvers, labs |
| Cloud & paid only | Offline mode, free keys optional |

**Common limitations found:** research systems summarise scientific papers rather than lecture decks, and they trade faithfulness for brevity; commercial tools produce attractive slides but cannot be checked against the source, ignore picture-only and equation slides, add no interaction and require a connection and a subscription. None of them redesigns the deck a teacher already uses — which is exactly the gap Learnova targets.

# TECHNICAL APPROACH — ARCHITECTURE

*Input → processing workflow → output; every AI step has an offline fallback*

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## System architecture and data flow

### INPUT

- **PowerPoint (.pptx)** — teacher's existing deck
- **Slide PDF** — exported lecture slides
- **Word / notes** — .docx, markdown, text
- **Typed topic** — a prompt or syllabus line
- **Teacher labs** — own HTML labs via API

*IR*

### INGEST

- **LibreOffice convert** — PPTX → PDF, cached by hash
- **Slide reader** — titles · bullet levels · tables
- **OCR decision** — Tesseract when text is reliable
- **Repair** — furniture, glued titles, split words
- **Colour DNA** — source palette → theme

*plan*

### ENGINES

- **Planner** — split · merge · agenda · glossary
- **Decision engine** — 27 detectors, reason + score
- **Layout engine** — glyph-measured, no broken words
- **Explainer solvers** — exact step-by-step (SymPy)
- **Quiz + lab selector** — checkpoints, one lab per topic

*calls*

### AI SERVICE

- **LLM router** — Gemini ×5 → Groq → NVIDIA
- **AI polish** — assertion titles, explanations
- **Topic diagrams** — validated SVG + HTML page
- **AI labs** — JS checked with node --check
- **Offline fallback** — rules, scenes, structure labs

*assets*

### OUTPUT

- **Editable PPTX** — native shapes, linked pictures
- **Web deck** — Reveal.js, reveals, overlays
- **Study pack** — flashcards · Anki · quiz bank
- **Share link** — view-only, no login
- **Deck tutor** — answers from the deck only

### DEPLOYMENT AND OPERATIONS · open-source / free tier

- **GitHub** — source · 755 tests
- **Vercel / Netlify** — React static build
- **Docker on Render** — FastAPI + LibreOffice + OCR
- **SQLite volume** — users · decks · caches
- **Monitoring** — health · audit · logs

**1 · Read faithfully** — every page, every figure kept

**2 · Decide with rules** — explainable visual per slide

**3 · Enrich with AI** — checked titles, diagrams, labs

**4 · Build and audit** — PPTX + web deck, no breaks

## Technologies to be used

React · Vite · Tailwind · Reveal.js · Python · FastAPI · SQLite · PyMuPDF · LibreOffice · Tesseract · Gemini · Groq · SymPy · pytest · GitHub · Docker

- **Frontend:** React 18 · Vite · Tailwind CSS · Reveal.js web deck · PWA (manifest + service worker)
- **Backend:** Python 3.12 · FastAPI · Pydantic · background job queue · SQLite · OTP auth
- **Documents:** python-pptx · PyMuPDF · LibreOffice headless · python-docx · Tesseract OCR
- **AI / NLP:** Gemini (pool of 5 keys, per-key pacing) · Groq · NVIDIA NIM · failover router
- **Maths & labs:** SymPy exact solvers · JavaScript lab engine · SVG via MuPDF
- **Quality & DevOps:** pytest (755) · layout audit · gold sets · GitHub · Docker

# TECHNICAL APPROACH — METHODOLOGY

*Nine stages from upload to audited deck, with the algorithm behind each*

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### 1 Convert
PPTX → PDF (LibreOffice), DOCX/text → Markdown IR; colour palette of the source kept.

### 2 Read slides
Furniture = text on ≥ max(3, 0.3n) pages; bullet level $\ell(x)$ = argmin $|x - a_k|$; glued titles split by DP.

### 3 OCR decision
$\gamma = |\text{ink} \setminus \text{text}| \,/\, |\text{ink}|$; OCR used only if conf ≥ 62, $\gamma \le 0.42$, < 15% symbol lines.

### 4 Plan
Split overloaded slides at logical boundaries; merge thin parts; agenda, glossary, quizzes.

### 5 Decide visual
27 detectors → (family, confidence, data, reason); $d^*$ = argmax c after exclusion rules.

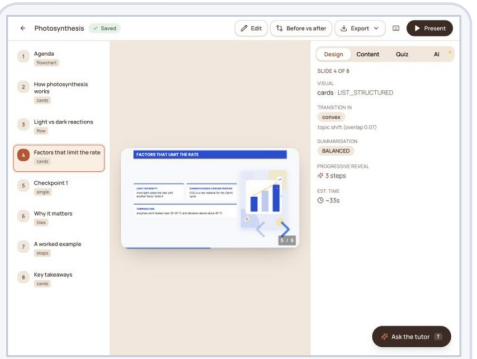
### 6 AI polish
Assertion titles, explanations, key terms — accepted only if grounded in the slide's words.

### 7 Illustrate
LLM SVG → whitelist + ≥ 6 shapes + render check → cache; else own figure / offline scene.

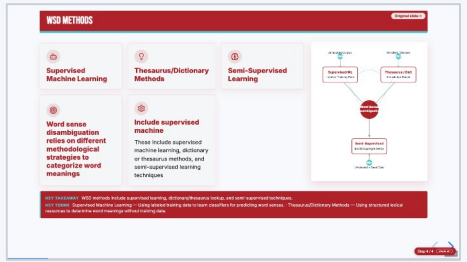
### 8 Lay out & build
$s^*$ = largest size that fits (≥ 11 pt); $\sigma = \min(1, W / w(\text{longest word}))$; PPTX + web deck.

### 9 Audit
Overflow, broken words, sparse slides; fact retention $R = |F_{src} \cap F_{deck}| \,/\, |F_{src}|$.

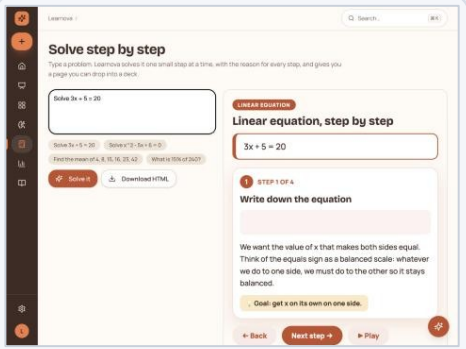
**WORKING PROTOTYPE · real screens from the running application**



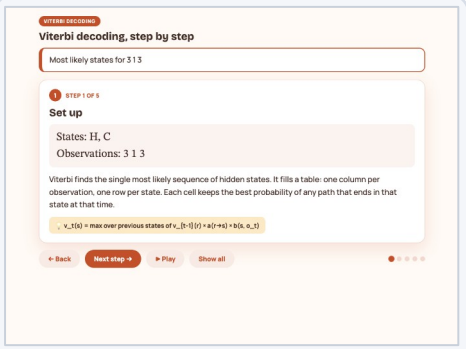
Deck preview — visual chosen and why



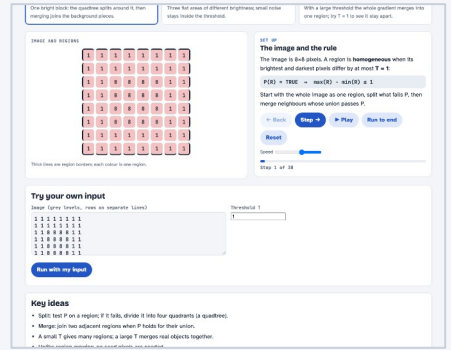
Web deck — AI diagram beside the content



Solve step by step



Viterbi explainer, exact



Split & Merge lab (IPCV)

# BEFORE vs AFTER

*The problems teachers and students faced — and what Learnova now achieves*

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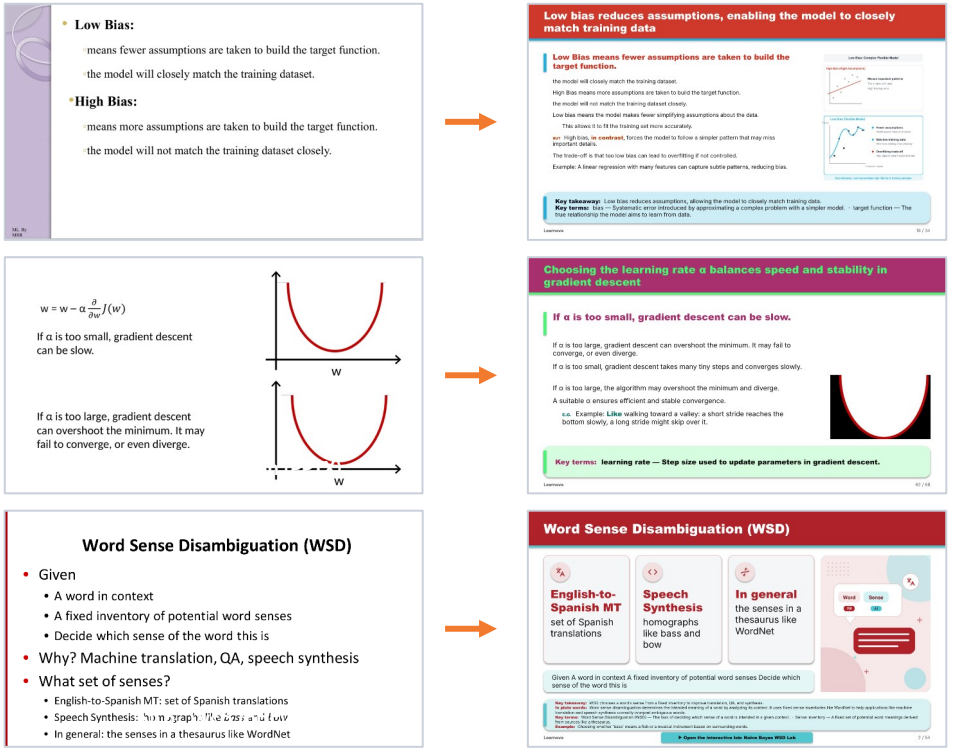
| Area | BEFORE — problem faced | AFTER — with Learnova |
|---|---|---|
| **Text-heavy slides** | ✘ 100+ words on a slide; students skim and miss points | ✔ One idea per slide, drawn as a visual at a readable density |
| **Pictures of equations** | ✘ Content locked in images; tools drop these slides | ✔ Kept full size as figures; OCR only when reliable |
| **Untitled / generic titles** | ✘ “Slide 9”, “Hypothesis (2/5)” says nothing | ✔ Assertion titles; 83 untitled slides → 0 |
| **No diagrams** | ✘ Plain text with nothing to explain the idea | ✔ Illustration on every topic slide + HTML link |
| **No interaction** | ✘ Solved problems shown as finished tables | ✔ Quizzes, step-by-step solvers, labs |
| **AI changes content** | ✘ Paraphrased, merged, invented facts | ✔ Verbatim + checked: 98–100% facts kept |
| **Redesign takes hours** | ✘ An evening per lecture, needs design skill | ✔ ≈10–60 s per deck, fully editable |
| **Needs internet / paid AI** | ✘ Cloud-only, subscriptions, quotas | ✔ Offline mode; 5 free keys rotate |

## Reading the comparison

The left column lists the problems we saw in our own courses: decks that are hard to read, half their content locked in pictures and no way for a student to check understanding. The right column is what Learnova does to the same material, measured on five real decks (two machine-learning PowerPoints and three NLP PDFs). The slide pairs show the same source slide before and after; the counters on the right sum the changes across all five decks.

The content itself does not change: fact retention stayed between 98% and 100% for every deck, and the layout audit found no overflowing text and no broken words.

**ORIGINAL → LEARNOVA**




**83 → 0**
untitled slides

**251**
slides drawn as visuals

**190**
linked illustrations

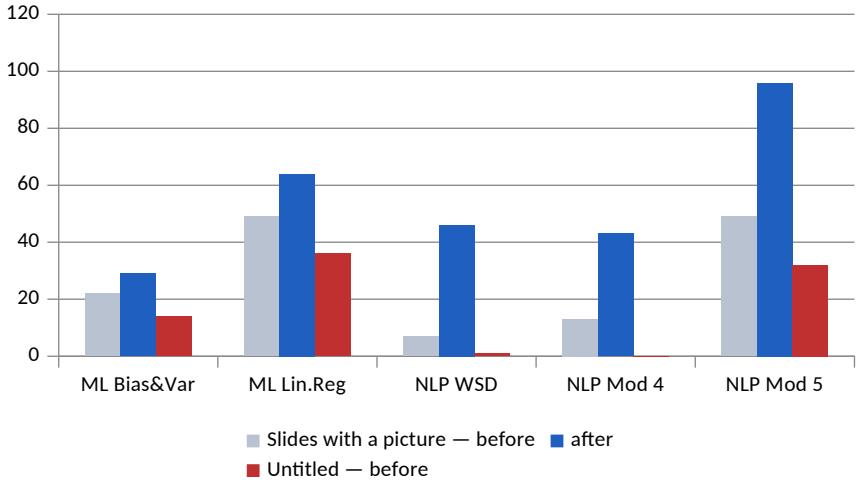
**64**
quizzes + labs added

# RESULTS, IMPACT & BENEFITS

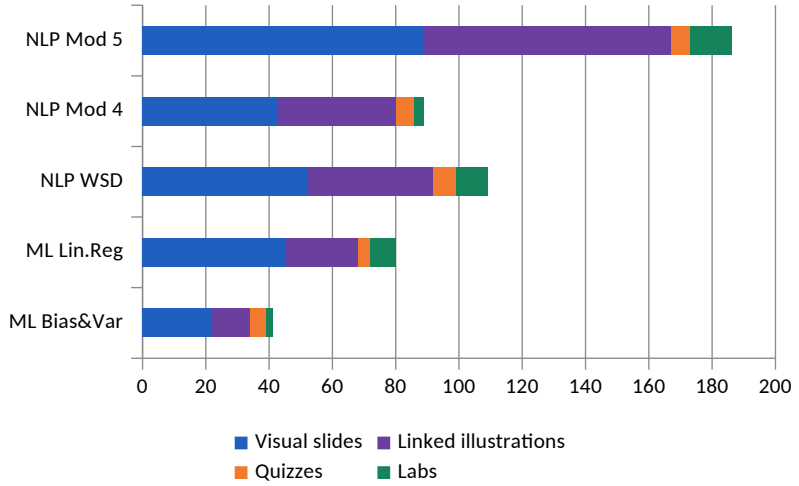
*Measured on real course material — not on promises*

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### Pictures and titles per deck




### Added by Learnova (all were 0 before)




**97%**
**visual chosen correctly**
101 gold slides, 3 sets

**98–100%**
**facts kept**
7 real decks

**0**
**broken words / overflow**
layout audit

**26/26**
**problems solved right**
held-out benchmark

**< 1 ms**
**decision per slide**
rule engine

**~35 s**
**216-page PDF → deck**
no API key

## Students

- ✔ One clear idea per slide with a picture
- ✔ Steps with reasons for every solved problem
- ✔ Quizzes, labs and flashcards to practise
- ✔ Revise on a phone, even offline

## Faculty

- ✔ Old decks modernised in about a minute
- ✔ Their own words and figures kept
- ✔ Editable PowerPoint, ready to teach
- ✔ Quizzes and labs with no extra work

## Institution

- ✔ Consistent, accessible course material
- ✔ No paid AI licence needed
- ✔ Supports flipped and remote classes
- ✔ Reusable lab library across subjects

## Society · SDG 4

- ✔ Quality education with low-cost tools
- ✔ Works where internet is unreliable
- ✔ Plain-language help for first-generation learners
- ✔ Open formats, easy to share

**Use cases:** a teacher uploads last year's PPT the night before a lecture and presents the redesigned deck; a student opens the shared web deck to revise with quizzes and labs; an IPCV class explores region growing and split-and-merge in the browser; a college without paid AI tools uses the offline mode. Next: a classroom study of learning outcomes with our own batch.

# FEASIBILITY & VIABILITY

*Running today on a laptop · pilot-ready for a department · every risk has a named safeguard*

**LEARNOVA · Group 1**
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### TECHNICAL
Already working end to end: FastAPI back end, React front end, 755 automated tests, layout audit. Each stage is a separate module, so features are added without rework.

### DATA
Input is the college's own lecture material (PPTX, PDF, DOCX). Tested on seven real decks of 15 to 216 slides from NLP, ML and IPCV.

### ECONOMIC
Free to run offline; Gemini free-tier keys (pool of 5) are optional. Hosting a department server costs less than a single paid AI-presentation licence per year.

### OPERATIONAL
Teachers keep control: every slide stays editable and every automatic choice shows its reason. Students need only a browser.

### SECURITY
OTP sign-in, hashed codes, rate limits; AI-written SVG and lab code are whitelisted and statically checked; share links are view-only.

## Challenges / risks → our safeguard

1. **Messy lecture PDFs**
   symbol bullets, glued titles, picture slides → page reader, DP title split, OCR decision
2. **AI hallucination**
   paraphrase, invented facts → grounded checks, verbatim source, SymPy numbers
3. **API quota / cost**
   free-tier limits → 5-key pool, per-key pacing, failover, offline mode, cache
4. **Broken layout**
   overflow, split words → glyph-measured fitting, 11 pt floor, continuation slides
5. **Slow large decks**
   216 pages, many diagrams → parallel generation, content-hash caching
6. **Unsafe generated code**
   scripts in SVG/labs → element whitelist, node --check, no network APIs
7. **Wrong visual choice**
   rule misfires → alternatives shown in the editor; slide stays editable

## Adoption path

**1 · Pilot**
AI & DS department: our own courses this semester, feedback from 3–4 faculty

**2 · Department**
shared server, deck library per teacher, labs library grows with faculty labs

**3 · College / LMS**
Moodle / Google Classroom export, regional languages, analytics

## Deployment plan

- **Front end:** static React build on Vercel / Netlify.
- **Back end:** Docker container (FastAPI + LibreOffice + Tesseract) on Render, Railway or Google Cloud Run with a persistent volume.
- **Config:** API URL for the front end; LEARNOVA_PUBLIC_URL so PowerPoint lab and illustration links resolve.
- **Why not all on Vercel:** long jobs, system binaries and on-disk storage do not fit serverless functions.

# RESEARCH & REFERENCES

*Papers we built on · tools we compared · our own publications and code*

**LEARNOVA · Group 1**
IPD Review · AI & DS, DJSCE

## SLIDE GENERATION


- Hu & Wan, PPSGen, IEEE TKDE 27(4):1085–1097, 2015
- Fu et al., DOC2PPT, AAAI 36:634–642, 2022
- Sun et al., D2S, NAACL 2021, pp. 1405–1418


## FAITHFULNESS


- Maynez et al., ACL 2020, pp. 1906–1919
- Ji et al., Survey of hallucination, ACM CSUR 55(12), 2023


## LEARNING SCIENCE


- Mayer, Multimedia Learning, CUP, 2009
- Sweller, Cognitive Science 12(2), 1988
- Garner & Alley, IJEE 29(6), 2013


## WORKED EXAMPLES & LABS


- Sweller & Cooper, Cogn. & Instr. 2(1), 1985
- Hundhausen et al., JVLC 13(3), 2002
- Wieman et al., PhET, Science 322, 2008


## LAYOUT & VISUALS


- Knuth & Plass, Breaking paragraphs into lines, SPE 11(11), 1981
- Mackinlay, APT, ACM TOG 5(2), 1986
- Moritz et al., Draco, IEEE TVCG 25(1), 2019


## NLP & OCR


- Jurafsky & Martin, Speech and Language Processing, 3rd ed. draft
- Smith, Tesseract OCR, ICDAR 2007
- Meurer et al., SymPy, PeerJ CS 3:e103, 2017


## TOOLS COMPARED

- Gamma (gamma.app)
- Microsoft Copilot in PowerPoint
- Gemini in Google Slides · Canva Magic Design

## OUR PAPERS

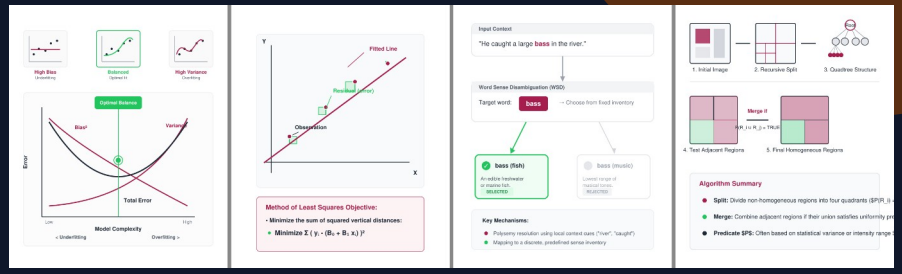
- IEEE conf. paper: Rebuilding lecture slide decks into visually structured, interactive presentations
- Springer LNCS paper: Computed step-by-step explainers and generated labs

**Scan to verify:** left — full source code, tests and demo decks (github.com/SDP42/LearnovaV2); right — both research papers (IEEE and Springer LNCS), gold sets and the measurement scripts used for every number on these slides. **Web resources:** revealjs.com · python-pptx · PyMuPDF · LibreOffice · Tesseract · ai.google.dev

# Thank You

## Questions, feedback — and a live demo of Learnova

Upload any lecture deck and watch it become an illustrated, interactive teaching deck.



Diagrams drawn by Learnova for real lecture slides


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Scan for

**code, papers & demo decks**

github.com/SDP42/LearnovaV2

**LEARNOVA · Read · Redesign · Illustrate · Engage · Deliver**